

From Specialization to Generalization: Instruction-tuned LLMs for Robust Harmful Content Mitigation

Lukas Edman^{1,2}

Daryna Dementieva^{1,2}

Alexander Fraser^{1,2,3}

¹School of Computation, Information and Technology, TU Munich

²Munich Center for Machine Learning

³Munich Data Science Institute

lukas.edman@tum.de, daryna.dementieva@tum.de

Abstract

Large language models (LLMs) demonstrate impressive performance across a wide range of general NLP tasks; however, their effectiveness in sensitive domains, such as hate speech detection, remains less clear. Prior studies comparing prompted LLMs with state-of-the-art encoder-based models (e.g., BERT variants (Roy et al., 2023; Dönmez et al., 2024)) have shown only marginal gains, suggesting that LLMs may not excel in hate speech detection or mitigation. In this work, we revisit this question through the lens of instruction tuning. By thoroughly unifying 36 English hate speech datasets spanning multiple labeling schemes, we fine-tune a generalist LLM, based on Qwen3 (Qwen Team, 2025), specifically for hate speech mitigation. Our results demonstrate not only state-of-the-art performance on in-domain benchmarks but also substantial improvements in cross-domain and cross-lingual generalization—areas where encoder-based specialist classifiers often struggle.

1 Introduction

Hate speech detection has been heavily studied, leading to the release of numerous datasets covering a wide spectrum of tasks, domains, languages, and annotation schemes (Mathew et al., 2021; Mandl et al., 2019; Cjadam et al., 2017) (see Table 1). Despite this diversity, progress in building *generalizable* hate speech models, which can cover a large variety of tasks related to hate speech as well as perform well on unseen domains, has been limited. Existing research largely treats each dataset or subtask in isolation, and little effort has been made to unify many available heterogeneous datasets to develop task-agnostic models with more robust performance across varied hate speech scenarios.

Consequently, task-specific encoder-based models (e.g., BERT variants) remain the dominant approach (Wang et al., 2020; Sai and Sharma, 2020),

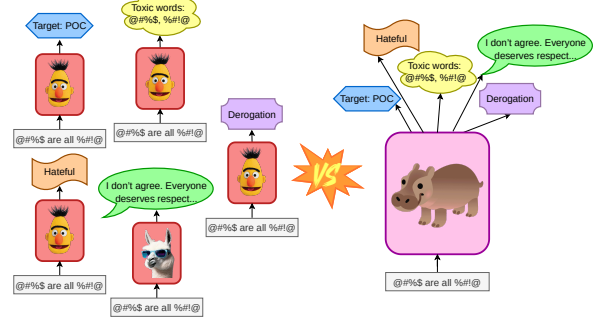


Figure 1: In this work, we would like to close the gap in understanding if instruction-tuned general-purpose LLMs (i.e., HIPPO, right) are more robust and precise detectors of hate speech than specially-tuned LMs (i.e., BERT and Llama, left).

as prompting large language models (LLMs) has not consistently yielded significantly superior results (Roy et al., 2023; Dönmez et al., 2024). At the same time, instruction tuning, one of the most promising methods for aligning LLMs to specialized domains, has not yet been systematically explored for hate speech mitigation.

To close this gap, firstly, we consolidate **36 heterogeneous datasets** in English into a single, instruction-formatted corpus spanning over 600k examples and 61 subtasks. Using this unified training dataset, we instruction-tune an LLM and name the resulting model **HIPPO** (Harmful Input, Positive and Productive Output, see Figure 1). We evaluate HIPPO across in-domain, out-of-domain, and cross-lingual settings. Our study aims to answer the following research questions:

RQ1: How does our instruction-tuned LLM perform compared to strong task-specific baselines for hate speech tasks in English?

RQ2: Does our instruction-tuned LLM outperform zero- or few-shot LLMs prompting for hate speech tasks?

RQ3: Can our instruction-tuned LLM generalize

to previously unseen hate speech domains and labels?

RQ4: To what extent can our English-only instruction-tuning dataset support cross-lingual generalization in LLMs?

Our results show that the instruction-tuned model achieves competitive, and often better, performance compared to best-performing encoder-based systems, while also exhibiting cross-task generalization. With a smaller version of the dataset to prime the model, we also see potentially higher cross-lingual performance, especially for generative tasks.

We release our code used for fine-tuning and the best-performing models online for public use.¹

2 Background

Hate Speech Definitions While a lot of work has been published in the domain of hate speech detection, very few computer science works mention explicitly what definition of hate or abusive speech they adopt in their studies for annotation or automatic processing (Arora et al., 2024; Rizwan et al., 2025). Thus, firstly, we specify which definitions we use in this study.

Hate speech mitigation lies under the term *digital violence*—an umbrella term that refers to words or actions that cause harm to an individual or a community. Our study focuses on digital violence, more specifically, expressed in a textual form. The study by Lewandowska-Tomaszczyk et al. (2023) categorizes harmful content as *offensive speeches*, including 17 sub-categories like *taboo*, *insulting*, *hate speech*, *harassment*, and *toxic*.

We adopt the definition of *hate speech* as abusive language targeting specific groups (Röttger et al., 2021), and *toxic speech* as texts that contain vulgar or profane language (Costa-jussà et al., 2022; Logacheva et al., 2022), but are not necessarily abusive or hateful. For many other labels that were then used in the instruction prompts, we refer to the original definitions of the datasets.

Hate Speech Mitigation Datasets Over the years, dozens of hate speech detection datasets have been introduced, reflecting a wide range of annotation strategies, domains, and languages (Poletto et al., 2021). The substantial efforts of the NLP community in this area are documented in the Hate Speech Data Catalog, which systematically tracks

the development of hate speech resources (Vidgen and Derczynski, 2020).²

Significant effort has been devoted to developing diverse hate speech detection and mitigation datasets, particularly for English. Common labeling schemes range from *binary* classification (e.g., hate vs. non-hate, toxic vs. non-toxic) to *multi-class* labels (e.g., hate speech vs. offensive language vs. neutral) and even *hierarchical* frameworks that further specify the subtype of hate once identified. One of the earliest curated resources, AMI18 (Fersini et al., 2018), was derived from Twitter and focused specifically on misogynistic content with a hierarchical labeling strategy. Then, many more datasets were gathered from various social networks like Twitter (Zampieri et al., 2019; Sachdeva et al., 2022), Gab (Mathew et al., 2021; Kirk et al., 2023), or Reddit (Vidgen et al., 2021; Qian et al., 2019). One of the most massive toxic speech classification datasets, Jigsaw (Cjadams et al., 2017), is based on Civil Comments and includes over 150k labeled samples.

Another notable dataset, HateCheck (Röttger et al., 2021), uses functionality tests to isolate specific cases in which hate speech manifests, allowing researchers to diagnose systematic model failures that may be obscured by aggregate benchmark metrics.

Beyond hate content *classification* datasets, several research directions have emerged that focus on more proactive hate speech mitigation through *text generation*. In particular, to reduce the publication of toxic or profane language—for example, in efforts to promote safer online environments for children—*text detoxification*, which aims to transform toxic text into neutral, has been proposed (Atwell et al., 2022; Dementieva et al., 2025a). For more severe cases of hate speech, approaches based on *counter-speech*, incorporating arguments and proactive dialogue, have also been explored (Yu et al., 2022; Bonaldi et al., 2022).

Together with English, many hate speech detection resources were created for other languages, for example, for Arabic and French (Ousidhoum et al., 2019), Spanish (Basile et al., 2019), Italian (Corazza et al., 2019), Portuguese (Fortuna et al., 2019), German (Jaki and Smedt, 2019), among others.

Hate Speech Supersets With many datasets already created for English and other languages, sev-

¹<https://github.com/TUM-NLP/HIPPO>

²<https://hatespeechdata.com>

Dataset	Attribution	Train	Test	Domain	Label Type	(Sub)Tasks
CAD	Vidgen et al. (2021)	13423			Multilabel	1
CAT-LARGE	Pavlopoulos et al. (2020)	19870			Binary	1
CONAN	Chung et al. (2019)	408		Synthetic	Generation	1
ConvAbuse	Cercas Curry et al. (2021)	4013		Chatbot	Hierarchical	4
DialoCONAN	Bonaldi et al. (2022)	1299		Synthetic	Generation	1
EAP	Vidgen et al. (2020)	19989			Multiclass	1
ETHOS	Mollas et al. (2022)	998			Hierarchical	4
FoxCom	Gao and Huang (2017)	1525		Fox News	Binary	1
GHC	Kennedy et al. (2022)	20610			Hierarchical	3
HSOL	Davidson et al. (2017)	24771			Multiclass	1
ImplicitHate	ElSherief et al. (2021)	6358			Multiclass	1
Intervene	Qian et al. (2019)	16628			Generation	1
LargeScaleAbuse	Founta et al. (2018)	44768			Multiclass	1
LargeScaleXDomain	Toraman et al. (2022)	68188			Multiclass	1
MeasuringHate	Sachdeva et al. (2022); Kennedy et al. (2020)	39511			Multiclass	1
Multitarget-CONAN	Fanton et al. (2021)	4055		Synthetic	Generation	2
NewsHate	Salminen et al. (2018)	3215			Hierarchical	2
ReligiousHate	Ramponi et al. (2022)	5746			Hierarchical	5
SlurCorpus	Kurrek et al. (2020)	39944			Multiclass	1
Stormfront	De Gibert et al. (2018)	10521		Stormfront	Binary	1
SWAD	Pamungkas et al. (2020a)	2567			Binary	1
ToxicCR	Sarker et al. (2023)	12767		Code review	Binary	1
TwitterCRT	Waseem and Hovy (2016)	12727			Binary	1
TwitterExpert	Waseem (2016)	6908			Multiclass	1
TwitterSA	Arkoshghalb (2019)	25825			Binary	1
USElect	Grimminger and Klinger (2021)	2400			Binary	1
AMI18	Fersini et al. (2018)	3998	1000		Hierarchical	3
EDOS	Kirk et al. (2023)	13998	2000		Hierarchical	3
HASOC19	Mandl et al. (2019)	5817	1153		Hierarchical	3
HatEval19	Basile et al. (2019)	8992	2971		Hierarchical	3
HateXplain	Mathew et al. (2021)	15348	1924		Hierarchical	3
Jigsaw	Cjadams et al. (2017)	159529	63978	CC	Multilabel	1
OffensEval20	Zampieri et al. (2019)	13187	5309		Hierarchical	3
ParaDetox	Logacheva et al. (2022)	11927	600		Generation	1
ToxicSpans	Pavlopoulos et al. (2022, 2021)	7931	2000	CC	Generation	1
Total		649761	80935			61

Table 1: List of datasets used and the number of examples, domain, and type of labels. Domains include Twitter () , Reddit () , Gab () , Facebook () , Wikipedia () , YouTube () , and Civil Comments (CC), among others.

eral works have attempted to unify them into a superset for ease of use. Antypas and Camacho-Collados (2023) made such an attempt with only *binary* labels 13 English hate speech detection datasets. Then, they performed a cross-dataset performance study utilizing mostly BERT-like or smaller LMs. Finally, MetaHate (Piot et al., 2024) aggregates 36 English datasets into one superset. These datasets either had binarized labels or multiclass labels that were subsequently binarized. This contrasts from our approach, where we do not substantially change the content of any labels, and additionally include generative tasks which are not trivial or possible to binarize.

Hate Speech Detection with Transformer-based Encoders Across numerous hate speech detection benchmarks, both in English and in other languages, state-of-the-art performance is predominantly driven by Transformer-based encoder models such as BERT (Devlin et al., 2019) and RoBERTa (Liu et al., 2019), including their multilingual variants. To further enhance domain-specific performance, HateBERT (Caselli et al., 2021) was introduced, extending BERT through

additional Masked Language Modeling on one million posts from banned Reddit communities.

Correspondingly, for datasets such as AMI18 and HASOC19, the strongest results were achieved using BERT and mBERT-based systems (Muti and Barrón-Cedeño, 2022; Mishra and Mishra, 2019). In the OffensEval20 shared task, the winning solution employed a RoBERTa-based architecture (Wiedemann et al., 2020). Likewise, for toxicity detection in the Jigsaw dataset, one of the top-performing and most widely adopted models (Logacheva et al., 2022) is a fine-tuned RoBERTa. Overall, BERT-style encoders continue to dominate performance in the hate speech detection domain.

Hate Speech Detection with LLMs With LLMs’ appearance, especially with additional human and safety alignment (Naveed et al., 2025), many experiments have been done to exploit such models for hate speech detection and automatic moderation.

Zhu et al. (2025) reports low agreement between LLM outputs and human labels, while Li et al. (2024) finds that LLMs are more reliable at identifying non-hateful content. Huang et al. (2023) studies LLM-generated explanations for im-

PLICIT hate, and Roy et al. (2023) shows that adding target-specific context improves prompting performance. Dönmez et al. (2024) evaluate a range of open- and closed-source LLMs prompting for hate speech and microaggression detection, showing that some models perform competitively. More advanced prompting techniques, like prompting with hate speech definitions or chain-of-thought prompting, showed promising results for real-world hate speech moderation (Guo et al., 2023). At the same time, prompting LLMs can lead to various undesirable and unstable outputs like hallucinations or refusal to process an input sample (Huang, 2025).

Several attempts have been made to fine-tune LLMs for hate speech detection tasks. Sen et al. (2024) tuned several 1B models with LoRA on two datasets. A comparison between prompting and fine-tuning of LLMs for sexism detection was performed in (Pan et al., 2024). Finally, Nasir et al. (2025) used LLMs’ hidden states as embeddings to fine-tune a smaller classifier. Nevertheless, none of the work explored massive tuning and generalization of LLMs to several hate speech tasks with different label schemes, hierarchies, and label types.

3 Methodology

We first describe our methodology for unifying datasets and converting them into the instructions, followed by our strategy for training on the unified dataset.

3.1 Data Collation

We source several English datasets, with the aim of including a diverse set of label types and domains. The resulting compilation is a superset of over 600k examples, shown in Table 1. Additionally, we also report all corresponding licenses of the datasets in Appendix D with several datasets samples examples in Appendix E. We divide the task types into 4 distinct categories:

- Binary classification
- Multiclass classification
- Multilabel classification
- Generation

Datasets labeled as ‘hierarchical’ include multiple tasks, where the main task is typically a binary classification, such as ‘hateful/not’, and the subsequent tasks only apply to those labeled as ‘hateful’.

Our test includes 9 of these datasets, which we selected primarily due to the reproducibility of their

train/test split, allowing for fair comparison. The resulting test superset includes tasks from each of our 4 categories.

Deduplication As many of the datasets are sourced from the same platforms, we deduplicate our training set to improve the quality. We first deduplicate based on the text/label pair, keeping only 1 copy. For examples with conflicting labels, we distinguish whether the examples are from the same dataset or not. If they are from the same dataset (as was largely the case with ToxiCR), we remove both copies. For cross-dataset duplicates, we temporarily binarize labels, and if they are still different, we keep the example with the ‘hateful or offensive’ label. This only affected 75 (0.01%) examples. In total, 3.73% (25147 examples) of the dataset is removed via deduplication. The final size of the dataset after deduplication is 649761, as presented in Table 1. Full statistics of the appeared labels in the final aggregated dataset is presented in Appendix C.

3.2 Instruction Construction

Our general approach is to convert every individual dataset into a conversational format, with user prompts and assistant answers. Our user prompt depends on the task, but all start with the generic message, followed by the input text, followed by a task-specific message, as shown below for binary classification.

User: You are an expert on hate speech, and you are tasked with answering questions about the following text. Respond only with the answer, no other text: {text}

Should the text be classified as {category}? Answer with ‘yes’ or ‘no’.

Assistant: {yes / no}

The category could be “hateful”, for example. For datasets with a hierarchical structure, we pose the questions in a back-and-forth conversational manner. For example, continuing from the previous example (assuming the answer is ‘yes’):

User: Which category does the text belong to? Categories: {categories}

Assistant: {category_choice}

This prompt is used for multiclass tasks. For the multiclass and multilabel tasks, we also prepend each category with letters (‘A. ’, ‘B. ’, ‘C. ’, etc.) so

Dataset	Type	HIPPO	Indiv.	GPT5	Best	Best Arch.	Best Source
AMI18 - A	Binary	81.0	74.8	70.9	71.0	BERT	Muti and Barrón-Cedeño (2022)
AMI18 - B	Binary x2	73.4	64.3	44.4	42.9	BERT	Pamungkas et al. (2020b)
EDOS - A	Binary	83.7	85.0	73.0	88.4	PaLM	Sorensen et al. (2023)
EDOS - B	Multiclass	71.2	65.5	41.9	73.3	PaLM	Sorensen et al. (2023)
EDOS - C	Multiclass	51.8	47.5	41.0	56.1	DeBERTa-v3	Zhou (2023)
HASOC19 - A	Binary	76.7	79.0	72.7	79.5	RoBERTa [†]	Kovács et al. (2021)
HASOC19 - B	Multiclass	58.5	56.7	54.8	54.5	BERT	Mishra and Mishra (2019)
HASOC19 - C	Binary	52.8	52.3	58.0	51.1	BERT	Mishra and Mishra (2019)
HatEval19 - A	Binary	56.8	55.7	66.8	70.8	Flan-UL2	Zhang et al. (2024)
HatEval19 - B	Binary x2	80.0	77.8	35.0	62.0	DistilBERT [†]	Atapattu et al. (2020)
HateXplain	Multiclass	69.5	69.1	40.8	69.9	BERT	Kim et al. (2022)
Jigsaw	Multilabel	80.7	80.2	61.5	62.5	BERT [†]	Mazari et al. (2024)
OffensEval20 - A	Binary	92.8	90.6	89.7	92.0	RoBERTa [†]	Wiedemann et al. (2020)
OffensEval20 - B	Binary	66.2	62.3	67.1	74.6	ERNIE 2.0 [†]	Wang et al. (2020)
OffensEval20 - C	Multiclass	70.3	67.0	60.7	71.5	ERNIE 2.0 [†]	Wang et al. (2020)
ParaDetox	Generation	64.8	65.2	35.3	74.2	DeepSeek	Hosseinbeigi et al. (2025)
ToxicSpans	Generation	64.3	63.1	25.9	70.8	BERT [†]	Zhu et al. (2021)
Average		70.3	68.0	55.3	68.5		

Table 2: Comparison of our unified model (HIPPO, based on Qwen3-4B-Instruct-2507) to individually trained models (Indiv.), GPT5-mini, and to the best performing models in literature. **Bold** values denote the best results per dataset. [†] denotes ensembles.

that the label for these tasks is simply the letter (or letters for multilabel). We found that this increases performance substantially over generating the exact names of the classes. For the multilabel case, the label is a comma-separated list of the present labels. We also consistently order the labels alphabetically, so that the model does not receive conflicting signals during training.

For generative tasks, there is no shared prompt, as every generative dataset’s task or tasks are unique. For example, for ToxicSpans, which we treat as a generative task, has the following prompt:

User: Identify the part(s) of the given text that is/are considered toxic. If there are multiple spans, separate them with semicolons. Write N/A if the text is not toxic.

Assistant: {span}

Other datasets, such as DialoCONAN, are even more open-ended, as the goal is to model a hate speech expert having a counter-narrative with someone saying harmful things. The conversational format naturally suits this type of dataset:

User: {narrative_1}

Have a dialogue with the author of the original text. Provide and maintain a counter-narrative throughout the dialogue.

Assistant: {counter_narrative_1}

User: {narrative_2}

Assistant: {counter_narrative_2}

...

User: {narrative_n}

Assistant: {counter_narrative_n}

We include an exhaustive list of our prompts in Appendix A.

3.3 Model Selection

We experiment with various LLMs of 4 billion parameters, including Qwen3 (Qwen Team, 2025), Llama3 (Dubey et al., 2024), and Phi4 (Abdin et al., 2024). We show the best-performing model—**HIPPO** based on **Qwen3-4B-Instruct-2507**, in Results (§4)—with the rest of LLMs in Appendix F. We use a smaller model with 4 billion parameters for our main experiments to limit ecological impact as well as to demonstrate the utility of instruction tuning for hate speech already from this size of computationally-accessible models. Nevertheless, we also finetune a larger (32B) and even smaller (0.6B) model to gauge the effect of model size.

3.4 Training

Training follows a typical instruction tuning setup. We apply QLoRA (Dettmers et al., 2023) to enable efficient fine-tuning, and we train the model only on the outputs of the assistant. Training on an Nvidia H100 GPU, the 0.6B model takes around 10 hours, the 4B models take around 1 day, and the 32B

model takes around 4 days. Additional training hyperparameters can be found in Appendix B.

4 Results

We go through the results as they relate to the research questions one-by-one. The scores shown in the following tables and figures are macro-F1, with some exceptions:

- AMI18 task B and HatEval19 task B are both composed of 2 subtasks, with their F1 scores averaged.
- ToxicSpans uses a character-based F1 score based on the overlap of the predicted span and label span.
- Jigsaw uses an average F1 score per-label.
- ParaDetox uses the joint scoring method from Dementieva et al. (2025b) that combines style-transfer accuracy, content preservation, and fluency.

More detailed results of HIPPO with confusion matrices for the classification-based test datasets are presented in Appendix G.

4.1 Comparison to State-of-the-art

RQ1: How does our instruction-tuned LLM perform compared to strong task-specific baselines for hate speech tasks in English?

Table 2 shows our main results on our test superset. We compare the performance of our unified training to training on each individual training split of each test set, finding that indeed our unified training brings a benefit in 14 out of 17 cases. Moreover, our unified approach outperforms a strong closed-source model, GPT5-mini.

Compared to the best-known models in the literature, the unified training approach wins on 7 of 17 tasks and achieves higher average performance. Notably, many state-of-the-art systems rely on task-specific ensembling, whereas we use no task-specific methods beyond prompting. In addition, competing non-BERT models often have substantially larger parameter counts (i.e., Flan-UL2 at 20B, PaLM at 62B, and DeepSeek at 671B), while our model is based on Qwen3-4B. As shown in Appendix G, our single model also generalizes across diverse tasks and label granularities, enabling a unified approach to hate speech mitigation.

In Table 3, we show that a 32B parameter model has even better performance overall, winning in

Dataset	0.6B	4B	32B	Best
AMI18 - A	28.1	81.0	82.9	71.0
AMI18 - B	26.0	73.4	75.5	42.9
EDOS - A	23.5	83.7	87.0	88.4
EDOS - B	42.0	71.2	74.7	73.3
EDOS - C	27.5	51.8	57.9	56.1
HASOC19 - A	22.2	76.7	76.6	79.5
HASOC19 - B	31.6	58.5	58.0	54.5
HASOC19 - C	47.6	52.8	52.7	51.1
HatEval19 - A	20.1	56.8	59.8	70.8
HatEval19 - B	41.0	80.0	79.8	62.0
HateXplain	16.1	69.5	70.8	69.9
Jigsaw	53.7	80.7	80.7	62.5
OffensEval20 - A	27.8	92.8	92.5	92.0
OffensEval20 - B	39.9	66.2	65.2	74.6
OffensEval20 - C	21.1	70.3	75.4	71.5
ParaDetox	10.3	64.8	65.0	74.2
ToxicSpans	21.9	64.3	64.2	70.8
Average	29.4	70.3	71.7	68.5

Table 3: Effectiveness of instruction tuning on varying model sizes, compared to the SOTA. **Blue** values indicate scores that outperform the SOTA. **Bold** values denote the best results per dataset.

11 of 17 cases versus the state-of-the-art. This indicates that the instruction tuning setup is even more effective at larger model sizes. At smaller model sizes, it appears ineffective. The 0.6B model performs particularly poorly, worse than random chance in many circumstances. This is because we do not constrain generation, so the model frequently does not output ‘yes’ or ‘no’ for binary classification tasks, for example.

Overall, the results show that from a unified, single-step training of a modest-sized LLM of at least 4 billion parameters, one can achieve performance competitive with several highly-tuned, task-specific models. We continue the further comparison of our HIPPO based on 4B-sized Qwen3.

4.2 Prompting and Cross-task Transfer

RQ2: Does our instruction-tuned LLM outperform zero- or few-shot LLMs prompting for hate speech tasks?

In Table 4, we first compare our fully-trained 4B model (i.e., HIPPO in Table 2) to zero-shot prompting the same base model. Our zero-shot prompts are identical to the instructions used in training. We additionally post-processed the zero-shot outputs to ensure variations of the correct answer (e.g., ‘yes’ versus ‘Yes.’) do not negatively affect the results.

We do not include results for few-shot prompting, as we were unable to formulate a few-shot

Dataset	Zero-shot	LOO	Full
AMI18 - A	59.8	78.4	81.0
AMI18 - B [†]	41.6	42.1	73.4
EDOS - A	66.7	76.0	83.7
EDOS - B [‡]	37.9	43.4	71.2
EDOS - C [‡]	20.2	27.6	51.8
HASOC19 - A	71.5	70.3	76.7
HASOC19 - B	31.2	52.0	58.5
HASOC19 - C	63.4	44.0	52.8
HatEval19 - A	58.1	52.5	56.8
HatEval19 - B [†]	33.3	29.3	80.0
HateXplain	46.8	60.1	69.5
Jigsaw [‡]	59.2	70.1	80.7
OffensEval20 - A	80.8	88.8	92.8
OffensEval20 - B	72.8	66.2	66.2
OffensEval20 - C	43.0	69.8	70.3
ParaDetox [‡]	48.0	46.4	64.8
ToxicSpans	22.1	42.3	64.3
Average	50.4	56.4	70.3

Table 4: Comparison of zero-shot to leave-one-out (LOO) and full HIPPO (i.e., unified) training. [‡] denotes tasks that do not have equivalent tasks in training of Qwen3-4B. [†] denotes partial equivalence: AMI18-B and HatEval19-B are composed of 2 tasks, both of which include 1 task that has no similar task in training. **Blue** values indicate scores that outperform the zero-shot baseline. **Bold** values denote the best results per dataset.

prompt that outperformed, or even matched performance of the zero-shot prompt. This has been previously observed in other works on prompting for hate speech detection, where few-shot prompting performed worse than zero-shot (Guo et al., 2023), or its benefits were inconsistent across languages (Ghorbanpour et al., 2025).

Nevertheless, we see that the fine-tuned model clearly outperforms our zero-shot prompt. While the prompts used are likely suboptimal, finding the optimal prompt for each task is time-consuming, likely more so than simply instruction tuning the model.

Concerning the leave-one-out (LOO) column in Table 4, we turn to our third research question:

RQ3: *Can our instruction-tuned LLM generalize to previously unseen hate speech domains and labels?*

To answer this, we train 9 additional models for each test dataset, each with the dataset’s corresponding training split left out.

As we can see, the results are generally better than the zero-shot performance, showing a clear cross-task learning capability. Of the 17 tasks, only

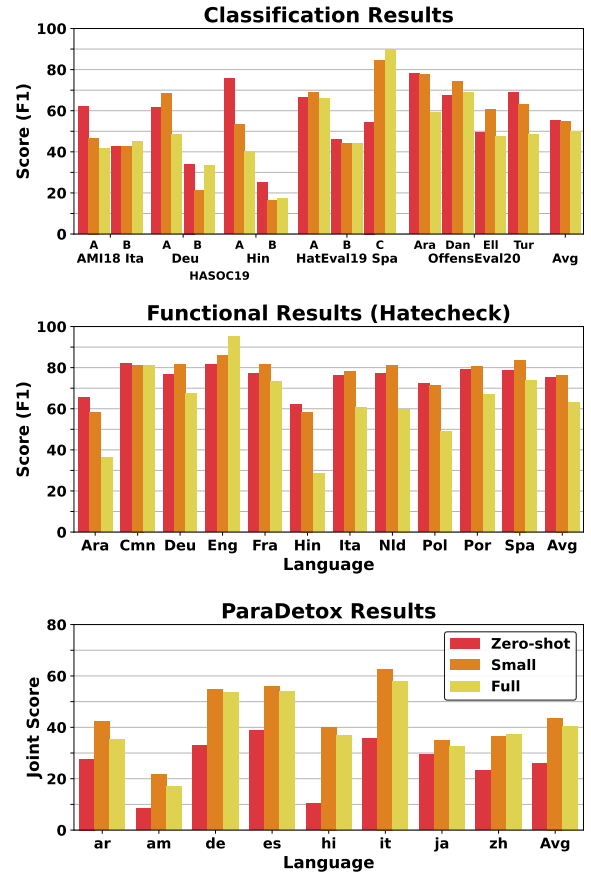


Figure 2: Cross-lingual and functional performance of the zero-shot, minimally-trained, and fully-trained HIPPO. Note: Unlike for English, HASOC19 does not have a Task C for German and Hindi.

6 have a decreased performance: HatEval19-A and B, HASOC19-A and C, OffensEval20-B, and ParaDetox.

Of the 6 tasks, HatEval19-B, HASOC19-C, and OffensEval20-B all have the same goal of determining whether a text is targeted or not. The issue may be due to dissimilar annotation guidelines or collection methods, as HatEval19 has a near-balanced training set (51% targeted, 49% untargeted), while HASOC19 and OffensEval20 have very lopsided distributions (85/15 and 88/12, respectively).

4.3 Cross-lingual Transfer

RQ4: *To what extent can our English-only instruction-tuning dataset support cross-lingual generalization in LLMs?*

While we focus on English for this work, high-quality hate speech processing is necessary for most languages. And since our base Qwen model has been pretrained multilingually, we therefore test whether our finetuning has damaged or helped

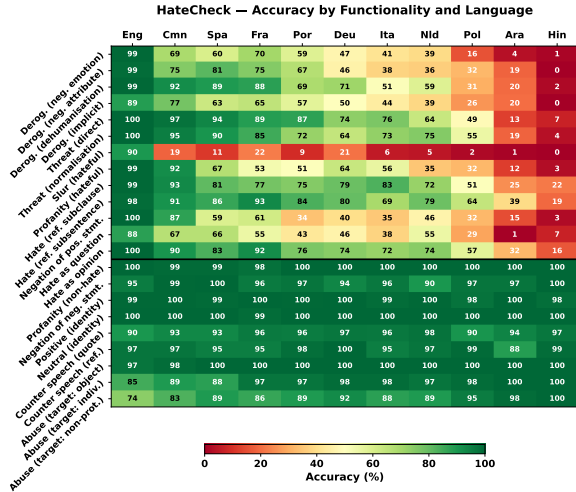


Figure 3: Performance of the fully-trained HIPPO on all languages (ordered by performance) of HateCheck, per functional task. Above the black line are hateful functionalities, below are non-hateful. Only functionalities shared across all languages are shown.

its non-English hate speech processing. We compare our zero-shot model with a fully-finetuned model, as well as an additional model trained on only 1k examples from each dataset. This smaller training set is intended to expose the model to the domain while not overfitting to English. We evaluate along 3 separate axes: classification, generation, and functional performance. For testing, we keep the prompts in English and only swap out the text and additional task information, such as class names. We show results in Figure 2.

In general, we observe that the fully trained model’s performance drops on both classification and functional tasks compared to zero-shot. As shown in Figure 3, this degradation varies by functionality in HateCheck. Arabic and Hindi perform worst, likely due to their distinct scripts, while Chinese performs better, possibly reflecting stronger representation in pretraining data. Performance on slurs is notably poor, consistent with their language-specific nature. In contrast, performance on non-hateful content remains nearly perfect across languages, suggesting a greater familiarity with non-toxic text from pretraining.

Returning to Figure 2, ParaDetox performance improves over zero-shot, which is unexpected. However, qualitative analysis shows the zero-shot model often translates inputs into English despite no instruction to do so, lowering its scores.

For the minimally trained model, performance is competitive on classification tasks, generally

higher on HateCheck, and substantially better on ParaDetox. This suggests that even limited English-only instruction tuning can yield positive cross-lingual transfer across diverse tasks.

There are many non-English hate speech training datasets available, so for truly better multilingual performance, we expect it would be highly beneficial to incorporate those into training. We leave this for future work. At the same time, we can see how the model is capable not only of classification, but for generation tasks already for multiple languages, which can serve as a strong out-of-the-box baseline.

4.4 Further Discussion

We additionally discuss two topics which may have a considerable impact on our models’ performances, label disagreement and data contamination, in Appendix H.

5 Conclusion

As hate speech can rapidly evolve in order to evade content moderation, the ability to perform generalizable hate speech mitigation has become increasingly important for practitioners who wish to curb the use of hate speech. To advance this goal, we introduced a unified instruction-style carefully curated training corpus constructed from 36 heterogeneous datasets, incorporating a wide range of harmful content classification tasks as well as several proactive speech generation tasks. We further evaluated multiple LLM families and sizes, and presented HIPPO, a model based on the Qwen3-4B variant, as an effective generalist solution.

We demonstrated that our generalist model performs at least on par with or significantly outperforms prior task-specific state-of-the-art BERT-style models, as well as prompt-based LLM approaches. We also saw a distinct increase in performance of our model on held-out sets, indicating an ability to generalize to new tasks which is quite impossible for BERT-based specialists. Finally, we showed that this generalization also extends to some multilingual tasks, although further training on multilingual data would likely benefit non-English languages further. We believe that our findings show the effectiveness of obtaining LLMs-based generalists for more robust harmful speech detection and proactive moderation across various domains.

Limitations

Our work is limited to training on only English datasets. While we do not expect that the trends would be different in other languages, we cannot know for sure without further experimentation. Training on multiple languages at once is also something we do not test, and could be more promising for better cross-lingual performance. However, it is also possible that cultural differences result in conflicting annotation styles that hurt performance.

We also do not test on a large number of prompting styles. For the finetuned model, the prompting style is likely not very impactful due to the additional training. For the zero-shot model, it is much more impactful. Our zero-shot results should not be seen as optimal zero-shot performance, but rather as performance on a manually crafted yet unoptimized prompt. There is additional information that could improve a prompt, such as definitions or annotation guidelines. Due to the lack of availability of this information for many datasets, we do not include it in our prompt.

We additionally do not incorporate or test model reasoning for hate speech tasks. Our GPT5-mini results used the default reasoning effort, “medium”, however we did not test other effort levels. Reasoning has been shown to be beneficial for many tasks, however we are not aware of any hate speech data with annotated reasoning, and the quality of LLMs’ hate speech detection is not good enough to make use of synthetic reasoning data.

Ethics Statement

Automatic moderation has become a promising application of modern NLP-based models, including recent LLMs. While such technologies may raise concerns about censorship, our work is strictly positioned as an investigation into the capabilities and limitations of modern language technologies, with the explicit goal of improving user safety in online environments. For this reason, we release our code and models under the OpenRAIL-S license, which restricts AI-based solutions usage to responsible and socially beneficial applications. The licenses of all used (and, later, our published) resources can be found in Appendix D.

Nevertheless, there remains a risk of misuse for dangerous scenarios—aggressive censorship, jail-breaking of models for hate behavior, or even generation of more hateful content. The genera-

tion of hateful or toxic language can already be achieved through simpler ways—for example, by injecting profane expressions or by relying on existing text style transfer models. Despite this accessibility, there has been no evidence of a large-scale surge in synthetic hateful content. Companies and researchers instantly continuing to improve safe mechanisms in LLMs making it incredibly difficult to jail-break with simple toxic injections.

In addition, we emphasize that effective content moderation cannot rely solely on automated systems. Responsible deployment requires sustained user-centered research and careful consideration of social context. Decisions about moderation policies should remain with the communities themselves, ensuring that safety mechanisms reflect their values, norms, and expectations.

At the same time, our proposed generalist hate speech mitigation model can now enable platform moderators to define more flexible and fine-grained content moderation guidelines, supporting scalable and consistent hate speech detection across large and diverse online environments. Then, our model outputs should serve as a recommendation for further moderation decisions, not as the ultimate final answer. We truly believe that our findings will serve as a base for future robust harmful human-written or AI-generated content detection for a safe online environment.

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A All Prompts

Here we list all prompts used in training and testing. They are categorized first by task type, followed by specific prompts for individual datasets. We omit the generic message before the input text as well as newlines for brevity. All of the prompts are included in our GitHub repository.

A.1 Generic Prompts

Binary Classification

User: Should the text be classified as {category}? Answer with ‘yes’ or ‘no’.

Multiclass Classification

User: Which category does the text belong to? Categories: {categories}

The categories are a newline-separated list, with ‘A. ’, ‘B. ’, ‘C. ’, etc. added before each entry. For tasks where the category is a specific target, we use this prompt instead:

User: Which of the following groups does the text target? Groups: {categories}

Multilabel Classification

User: Which of the following categories does the text belong to? You can choose multiple, but keep the order of the categories. Categories: {categories}

The categories are a newline-separated list, with ‘A. ’, ‘B. ’, ‘C. ’, etc. added before each entry. For tasks where the category is a specific target (and can be multiple targets), we use this prompt instead:

User: Which of the following groups does the text target? You can choose multiple, but keep the order of the groups. Groups: {categories}

A.2 Specific Prompts

Counter Speech For CONAN and Multitarget-CONAN, we use:

User: Provide a counter-narrative for the text.

For DialoCONAN, we use:

User: Have a dialogue with the author of the original text. Provide and maintain a counter-narrative throughout the dialogue.

Ethos Ethos has two annotations which required prompts that did not fit into our generic prompting strategies.

User: Does the comment incite violence? Answer ‘yes’ or ‘no’.

User: Is the text targeting a specific individual (directed) or a group/class of people (generalized)? Answer ‘directed’ or ‘generalized’.

Gab Hate Corpus One of the tasks in the GHC is to annotate whether the hateful content is said in an implicit or explicit way.

User: Is the rhetoric in the text explicit or implicit? Answer with ‘explicit’ or ‘implicit’.

HateExplain Although we do not use it for testing, we include the task of identifying parts of the text that are hateful or offensive. This prompt is only given if the text was classified as such.

User: Identify all of the words in the text that could be used to justify the given classification. Write them in a space-separated list.

Implicit Hate For implicit hate, we only use one of the tasks, which is to infer the speaker’s meaning from their comment.

User: What is the implied statement of this post?

Intervene The intervention dataset includes examples on multiple lines, with each line labeled with an index. The task is to first identify which lines constitute hate speech, then to form an intervention. Each example had 3 possible interventions. The detection prompt is as follows:

User: Which posts or comments in this conversation are hate speech? Refer to their numbers, e.g. 1, 2, 3, or n/a if none of the posts or comments are hate speech.

The intervention prompts are as follows:

Parameter	Value
Batch size	64
Learning rate	1e-4
Weight decay	1e-3
Warmup ratio	1e-2
Epochs	3
LoRA rank	32
LoRA alpha	32
LoRA dropout	0
LoRA bias	None
Optimizer	AdamW 8-bit
Scheduler	Cosine
Quantization	4-bit NF4 double

Table 5: Hyperparameters used in training.

User: How would you respond to intervene? Limit your response to 150 characters.

User: Come up with another response to intervene.

User: Come up with a third response to intervene.

ParaDetox

User: Rephrase the text to make it less toxic and more neutral.

SWAD

User: Is the word marked in bold () used in an abusive manner?

ToxicSpans

User: Identify the part(s) of the given text that is/are considered toxic. If there are multiple spans, separate them with semicolons. Write N/A if the text is not toxic.

B Hyperparameters

In Table 5, we list the hyperparameters specified in training.

C Datasets Labels Statistics

Here, we give a general overview of the label distribution of the data. Figure 4 shows the top-level labels of all datasets, grouped into 5 categories. The ‘generative’ category consists of the datasets

with generative tasks: ToxicSpans, ParaDetox, ImplicitHate, Intervene, and the CONAN datasets. If these were discriminative tasks, the examples would largely fall under the toxic or hateful categories, as the tasks presume the text is toxic or hateful.

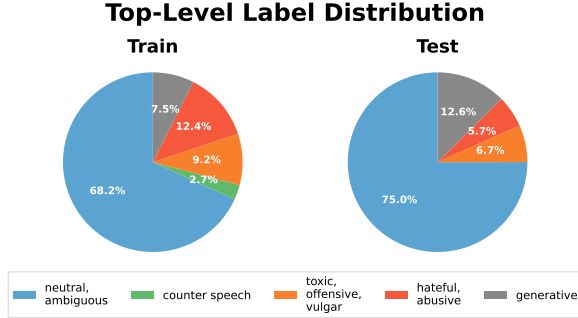


Figure 4: Top-level label distribution of all datasets.

In Figure 5, we break down the datasets with finer-grained labels, not included in Figure 4. We find that the finer-grained labels can be broken into two types: target-based and style-based labels. Target-based labels refer to labeling the target of the hateful or offensive speech. Style-based labels refer to the manner in which it is hateful or offensive. These labels are not mutually exclusive, nor are the labels within each category. We group similar labels (e.g., sexism, misogyny, and gender) for brevity; however, there may be overlap between some label groups. Overall, we see that target-based labels are more frequent in the training set, but the types are more evenly distributed in the test set. We also see that generic labels, such as whether an example is targeted or not, are more frequent. The large ‘other’ group mostly comes from the Gab Hate Corpus, in which examples were annotated for a variety of targets, so other could either refer to untargeted hate or targeted hate that does not fall into the included categories.

D Licensing of Resources

Below is an overview of the licenses associated with each resource used in this work (Table 6).

The licenses associated with the datasets utilized in this study are consistent with the intended use of conducting academic research on various NLP application for positive impact.

As all the licenses support public access to the data, we will opensource our final combined instruction training dataset with the most strict license that will support only research purposes that

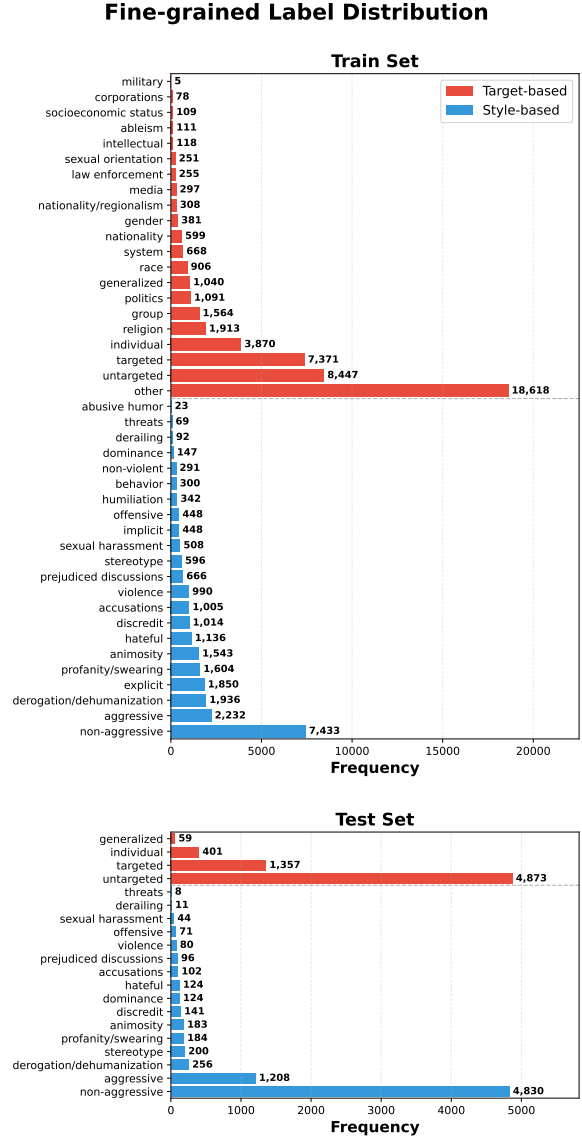


Figure 5: Fine-grained label distribution of the hierarchical datasets.

Resource	License	Homepage
CAD	CC BY 4.0	https://github.com/dongpng/cad_naac12021
CAT-LARGE	Apache-2.0	https://github.com/ipavlopoulos/context_toxicity
CONAN	Research Only	https://github.com/marcoguerini/CONAN
ConvAbuse	CC BY 4.0	https://github.com/amandacurry/convabuse
DialoCONAN	Research Only	https://github.com/marcoguerini/CONAN
EAP	CC BY 4.0	https://zenodo.org/records/3816667
ETHOS	GNU GPLv3	https://github.com/intelligence-csd-auth-gr/Ethos-Hate-Speech-Dataset
FoxCom	MIT	https://github.com/sjtuprog/fox-news-comments
GHC	CC BY 4.0	https://osf.io/edua3/overview
HSOL	MIT	https://huggingface.co/datasets/tdavidson/hate_speech_offensive
ImplicitHate	MIT	https://github.com/SALT-NLP/implicit-hate
Intervene	CC BY-NC 4.0	https://github.com/jing-qian/A-Benchmark-Dataset-for-Learning-to-Intervene-in-Online-Hate-Speech
LargeScaleAbuse	CC BY	https://github.com/ENCASEH2020/crowdfollower-platform
LargeScaleXDomain	CC BY-NC 4.0	https://github.com/avaapm/hatespeech
MeasuringHate	CC BY 4.0	https://huggingface.co/datasets/ucberkeley-dlab/measuring-hate-speech
Multitarget-CONAN	Research Only	https://github.com/marcoguerini/CONAN
NewsHate	CC BY	https://ojs.aaai.org/index.php/ICWSM/article/view/15028/14878
ReligiousHate	MIT	https://github.com/dhfbk/religious-hate-speech
SlurCorpus	MIT	https://github.com/networkdynamics/slur-corpus
Stormfront	CC BY-SA 3.0 ES	https://github.com/Vicomtech/hate-speech-dataset
SWAD	CC BY NC	https://aclanthology.org/2020.lrec-1.765.pdf
ToxicCR	GPL-3.0 license	https://github.com/WSU-SEAL/ToxicCR
TwitterCRT	CC BY 4.0	http://github.com/zeerakw/hatespeech
TwitterExpert	CC BY 4.0	http://github.com/zeerakw/hatespeech
TwitterSA	Unknown	https://www.kaggle.com/datasets/arkhoshghalb/twitter-sentiment-analysis-hatred-speech
USElect	CC BY 4.0	https://www.ims.uni-stuttgart.de/forschung/ressourcen/korpora/stance-hof/
AMI18	CC BY-NC 4.0	https://github.com/MIND-Lab/Automatic-Misogyny-Identification
EDOS	CC0-1.0	https://github.com/rewire-online/EDOS
HASOC19	Open Access	https://hasocfire.github.io/hasoc/2019/call_for_participation.html
HatEval19	CC BY 4.0	https://github.com/msang/hateval/tree/master/SemEval2019-Task5
HateXplain	CC BY 4.0	https://huggingface.co/datasets/Hate-speech-CNERG/hatexplain
Jigsaw	CC0-1.0	https://www.kaggle.com/competitions/jigsaw-unintended-bias-in-toxicity-classification/data
OffensEval20	CC BY 4.0	https://sites.google.com/site/offensevalsharedtask/offenseval-2019
ParaDetox	OpenRail++	https://huggingface.co/datasets/s-nlp/paradetox
ToxicSpans	CC0-1.0	https://huggingface.co/datasets/heegy/toxic-spans

Table 6: Overview of the licenses associated with each resource utilized in this work for experiments.

serve responsible usage of datasets and the model for social good.

E Datasets Examples

Here, we provide several examples of the texts and the corresponding labels from the used datasets for the classification task (see Table 7) as well as datasets used for the hate speech mitigation with generation tasks (see Table 8).

Warning: The examples are included solely to illustrate the nature of the dataset and are not intended to offend or target any individual or group.

F Model Comparison

In this section, we compare varying models to understand their effect. Our main model, based on Qwen3-4B-Instruct-2507, is a newer version of Qwen3, focused only on instruction tuning, rather than a mixture of instruction and reasoning data. We compare to the older Qwen3 model, simply named Qwen3-4B. We disable its reasoning capabilities. To better distinguish between the two, we refer to our main model as HIPPO and the older model version as Qwen3. We also compare to Llama3.2-3B-Instruct, and Phi4-Mini-Instruct in Table 9.

Here, we see that while HIPPO performs best, and both Qwen-based models perform better overall, the other models show similar performance, demonstrating the effectiveness of our training setup on multiple architectures.

G Extended results

In Figure 6, we show the confusion matrices for every classification-based task tested (for English) for our unified model.

Overall, our unified model demonstrates strong performance across a range of labeling schemes, quite accurately predicting binary categories from diverse domains—such as hate vs./non-hate, misogynous/non-misogynous, and offensive/non-offensive—with minimal label confusion. It also performs effectively on more fine-grained classification tasks, including AMI18-Task B2 and EDOS-Task C, which distinguish between categories such as slurs, attacks, threats, harassment, and others. At the same time, the model occasionally confuses targeted and untargeted content and, in some tasks, hateful language with just offensive ones. Nevertheless, these detailed per-label results indicate that a single LLM tuned on the heterogeneous harmful content detection tasks can

Dataset	Examples	Label
AMI18 SUBTASK 1	There are some wins that will make u go hysterical. Some makes you happy but same time you are sad	Non-misogyny
	Shut the f*ck up you f*cking monkey jew b*tch c*nt n*gger d*ck l*cking	Misogyny
	he thing is if a woman speaks out about rape on Nigerian twitter.The backlash would be shocking. It’s not good for the mind. Hence silence	Non-misogyny
EDOS SUBTASK 2	MRA stuff like men saying they wished they hadn’t had daughters, calling ALL American women wh*res, etc. Evidence if this?	Threats
	Is this ASD (anti sl*t defence) or what? How do I prevent this? Spend more time with her before and after.	Derogation
	Not our fault she chose to date an abusive sc*mbag.	Animosity
	Someone k*ll her, really, I’m not kidding, someone c*t her throat	Prejudiced Discussion
HATEVAL19 SUBTASK 1	And people still want to let them all in?!? #BuildTheDamn-Wall #NoDACA #ICE #LockThemAllUp #SendThemBack	Hateful
	It’s time for Africans to reevaluate their relations with #China.	Non-hateful
	Bitch shut up u a whore	Hateful

Table 7: Examples from a few classification datasets used in our work for experiments. We included multiple subtasks and domains, covering just high-level classification of hate/non-hate as well as other more fine-grained labels.

Dataset	Examples
TOXICSPANS	Input Text: your an id*ot, this is a tax based on a lie Toxic Span: id*ot
	Input Text: Have you walked around Denver lately , or Civic Center Park ? These poor homeless people are p*gs ! Toxic Span: p*gs
	Input Text: M*ron. Illegals have no Constitutional rights Toxic Span: M*ron
PARADETOX	Input Text: the police dont have time for this cr*p. Detoxified: The police have no time for this.
	Input Text: its a crock of sh*t , and you know it . Detoxified: It’s senseless, you know
	Input Text: i didnt vote for the liar Detoxified: I didn’t vote for him

Table 8: Examples from a few generation datasets used in our work for experiments. We included not only classification, but generation tasks showing that instruction-tuned model can be used for more variety of proactive ways for hate speech mitigation.

Dataset	Llama3.2	Phi4-mini	Qwen3	HIPPO
AMI18 - A	78.7	75.7	79.2	81.0
AMI18 - B	70.3	54.8	72.5	73.4
EDOS - A	83.5	83.5	84.6	83.7
EDOS - B	70.0	61.9	67.5	71.2
EDOS - C	43.8	41.4	47.5	51.8
HASOC19 - A	77.6	74.5	78.1	76.7
HASOC19 - B	62.2	60.4	56.6	58.5
HASOC19 - C	52.7	53.1	50.4	52.8
HatEval19 - A	55.2	56.9	57.6	56.8
HatEval19 - B	77.4	79.8	79.8	80.0
HateXplain	68.8	66.2	70.2	69.5
Jigsaw	80.6	79.9	80.1	80.7
OffensEval20 - A	92.6	93.1	92.6	92.8
OffensEval20 - B	65.9	73.0	69.0	66.2
OffensEval20 - C	70.9	54.7	72.0	70.3
ParaDetox	69.4	68.0	64.9	64.8
ToxicSpans	62.6	65.1	63.3	64.3
Average	69.5	67.2	69.8	70.3

Table 9: Comparison of model types. We can observe that Qwen3 family models (Qwen3 and HIPPO) perform better and more stable in comparison to other same-sized models from other families.

indeed predict beyond binary labels and generalize across multiple domains, task formulations, and categories granularities.

We compare the unified model against the individual models in Figure 7. From these, we can see that, the results on AMI18-B and EDOS-B improve with the unified model due to less confusion of the categories *discredit* and *dominance* in AMI18, and *derogation* and *animosity* in EDOS. There is a high overlap in *discredit* and *derogation* in their definition (as defined by the shared tasks), and similarly for *animosity* and *dominance*, so the unified model is able to benefit from the similarities of these two tasks, while the individual models are unable to do so.

H Further Discussion

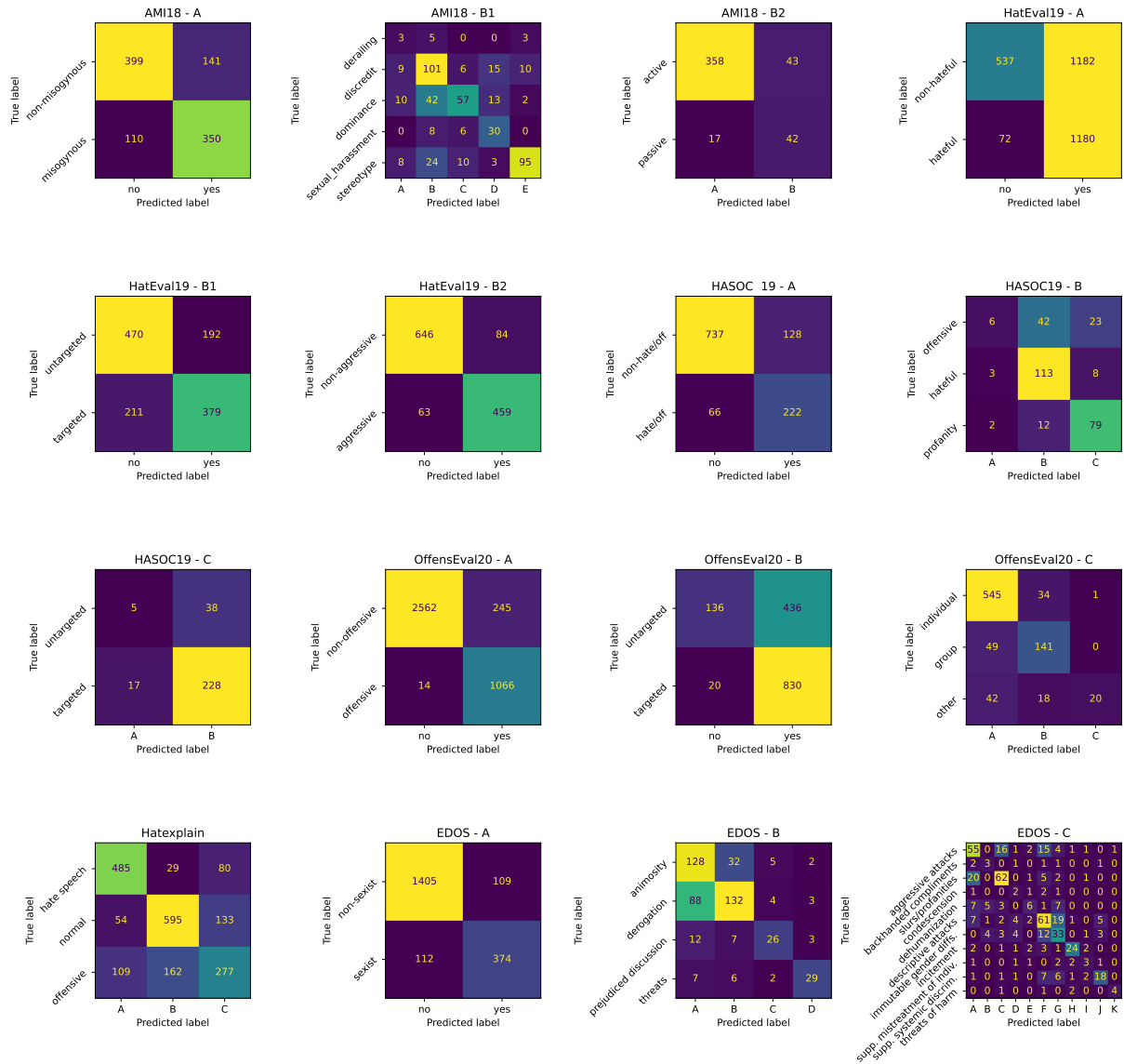
Label Disagreement Unifying hate speech datasets has an underlying issue: label disagreement. While we did handle this in our filtering step for exact duplicates, sentences with similar sentiment are more difficult to identify, especially in the context of hate speech, where subtle differences can lead to different classifications. Fortuna et al. (2020) noted that label disagreement is a major inhibitor in unifying datasets. Annotator guidelines are a contributing factor, with different definitions

of hate speech and inconsistent categorizations that do not overlap across datasets. Annotators themselves can also have disagreements that lead to inconsistencies within a dataset.

As we noted in our results in Section 4.2, differing distributions among datasets could be a result of different guidelines, and can hurt the cross-task performance if the model has not been trained on that specific dataset. We note that in the HatEval19-C task of determining whether a tweet is targeted, the guidelines specify it should target an individual only, not a group, while other datasets include groups (e.g., HASOC19).

Including definitions and guidelines within a prompt could be a solution to this, however it also makes it difficult to prevent overfitting. If the guidelines and definitions are always the same, it will become a signal for that specific dataset. A future approach may be to paraphrase these guidelines for increased variety. Nevertheless, our joint model outperforms individually trained models, so the benefits of joint training overall outweigh the drawbacks.

Data Contamination Data contamination is a widespread issue for closed-source LLMs, and it is difficult to determine the level of contamination, given that we have no access to the datasets



used in training. It is important to distinguish the differences in types of contamination, particularly *text contamination* versus *text-label contamination* (Jiang et al., 2024). As the majority of the datasets we use originate from public social media such as Twitter, it is highly likely that there is text contamination. However, this is also the case for a number of the best-reported models, as they are additionally trained on unlabeled data. For example, works on OffenseEval20 and EDOS included a large amount of unlabeled data for training, as the data was provided by the shared task organizers.

Concerning text-label contamination, it is more difficult to assess whether models have been contaminated in such a way. We are confident that at least our results on ParaDetox are not contaminated.

as the test sets were obtained from the organizers and are not freely available online.

To the extent that contamination affects our results, we can be confident that it does not affect RQs 2, 3, and 4, as these experiments all use the same base model. For RQ1, we can be sure that our unified method outperforms individual training, and our comparison to GPT5-mini should also be fair, as there is no reason to expect GPT5-mini to be less contaminated than Qwen3. Overall, while data contamination may inflate performance on some tasks, instruction tuning on hate speech data still remains a promising method for hate speech processing, clearly outperforming simple prompting, and demonstrating a level of cross-task generalization that is otherwise impossible with BERT-style

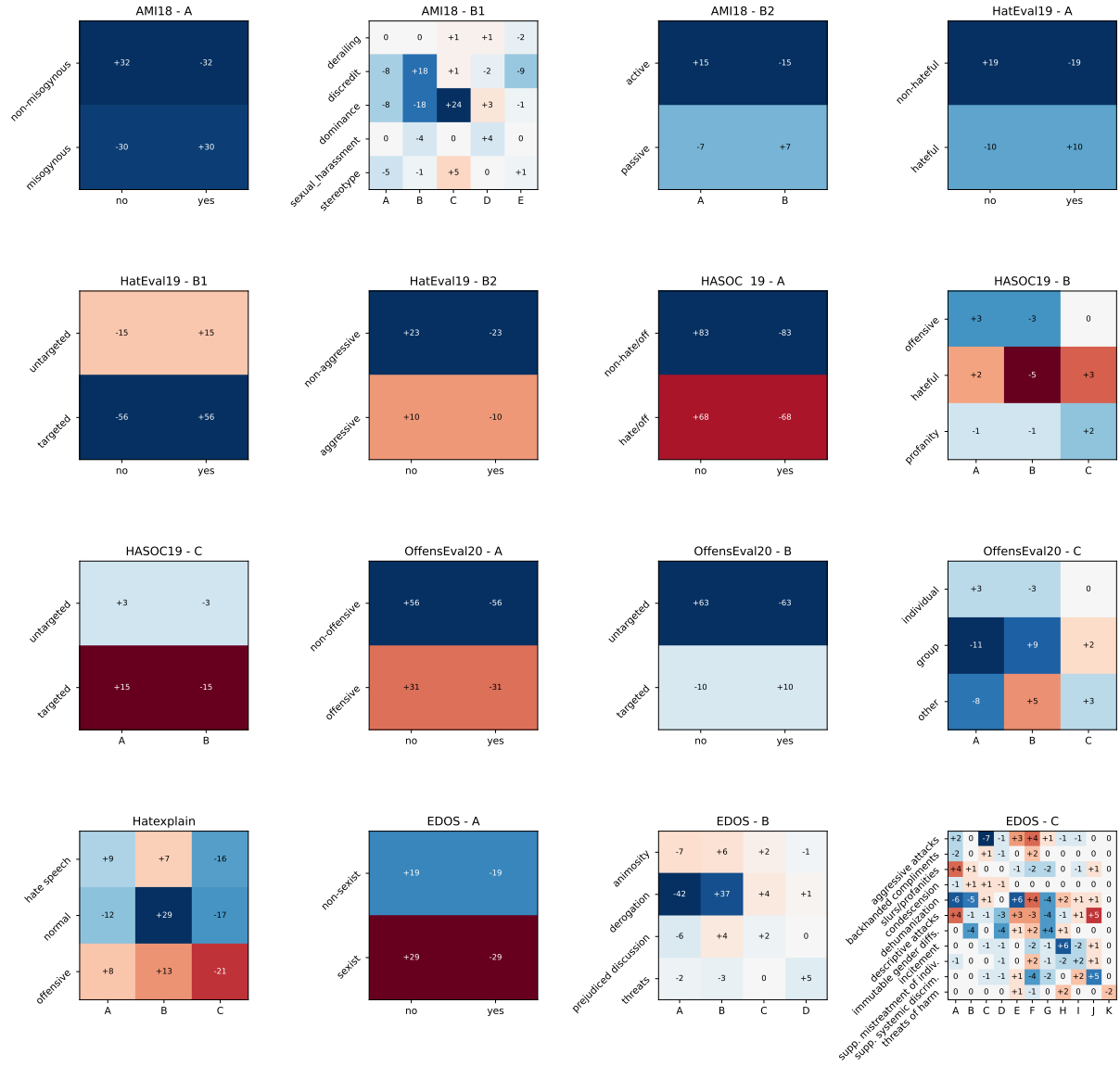


Figure 7: Difference of confusion matrices between the unified model and individual models. Blue indicates better performance for unified, red vice versa.

models.

I Usage of AI Assistants

In this paper, an AI assistant was utilized in the writing process. ChatGPT was used only for paraphrasing and checking the grammar in some paragraphs. During the development phase, Cursor was used for code completion and assistance.